

5G RAN

Intelligent Aggregate Scheduling (High-Frequency TDD) Feature Parameter Description

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Contents

1 Change History.....1

2 About This Document.....2

2.1 General Statements.....2

2.2 Features in This Document.....3

2.3 Differences.....3

3 Overview.....5

4 Uplink IAS.....6

4.1 Principles.....6

4.1.1 Uplink Packet Bundled Scheduling.....7

4.1.2 Packet-Length-based Scheduling Optimization.....8

4.1.3 Dual PRACH Periods.....8

4.1.4 PUCCH Receive Channel Adaptation.....9

4.1.5 Uplink IBLER Adaptation Optimization.....10

4.2 Network Analysis.....11

4.2.1 Benefits.....11

4.2.2 Impacts.....12

4.3 Requirements.....15

4.3.1 Licenses.....15

4.3.2 Functions.....15

4.3.2.1 Prerequisite Functions.....16

4.3.2.2 Mutually Exclusive Functions.....18

4.3.2.3 Function Impacts.....19

4.3.3 Hardware.....20

4.3.4 Cells.....20

4.4 Operation and Maintenance.....21

4.4.1 Data Configuration.....21

4.4.1.1 Data Preparation.....21

4.4.1.2 Using MML Commands.....23

4.4.1.3 Using the MAE-Deployment.....24

4.4.2 Activation Verification.....24

4.4.3 Network Monitoring.....24

5 Parameters.....26

6 Counters.....

28

7 Glossary.....

29

8 Reference Documents.....

30

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft A (2025-12-31)

This issue is the first release.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Trial Features

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial

features shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-101241	Uplink IAS	4 Uplink IAS

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Uplink IAS	None	4 Uplink IAS

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Uplink IAS	Packet-length-based scheduling optimization is supported in both high and low frequency bands. This document describes only how the packet-length-based scheduling optimization function works in high frequency bands. For details about how this function works in low frequency bands, see <i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>. The other functions are supported only in high frequency bands.	4 Uplink IAS

3 Overview

To better adapt to various user service behaviors and address issues of fast signal attenuation and limited beam resources in mmWave transmissions, mmWave intelligent aggregate scheduling (mmWave IAS) is introduced. **mmWave IAS optimizes scheduling policies, resource consumption policies, beam resource utilization, and system link performance**, thereby improving the user experience in and capacity of TDD mmWave cells.

4 Uplink IAS

4.1 Principles

The Uplink IAS feature is controlled by the **FR2_EXP_IMP_SW** option of the **NRDUCellPusch.PuschPerformanceSwitch** parameter. This feature includes the functions listed in [Table 4-1](#).

Table 4-1 Functions of Uplink IAS

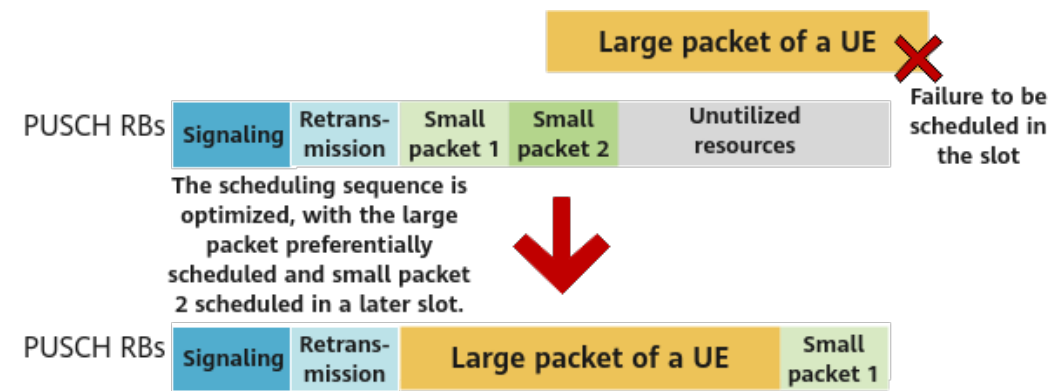
Function Name	Optimization Category	Principle Overview	Reference
Uplink packet bundled scheduling	Scheduling policy	This function adaptively adjusts the scheduling occasions for UEs running large-packet services to fully utilize beam resources and improve uplink physical resource block (PRB) usage.	4.1.1 Uplink Packet Bundled Scheduling
Packet-length-based scheduling optimization	Scheduling policy	Aimed at minimizing the total transmission delay of UEs to be scheduled, this function adaptively selects appropriate UEs for scheduling based on their traffic models and signal quality. This increases the average uplink UE throughput.	4.1.2 Packet-Length-based Scheduling Optimization

Function Name	Optimization Category	Principle Overview	Reference
Dual PRACH periods	Resource consumption policy	This function extends the physical random access channel (PRACH) period for contention-based random access (CBRA) and this allows the number of slots occupied by the PRACH within a given time segment to be reduced, thereby increasing the number of slots available for the physical uplink shared channel (PUSCH). In addition, this function enables the base station to receive preambles for contention-free random access (CFRA) in specific slots, further reducing the amount of time-frequency resources occupied by the PRACH.	4.1.3 Dual PRACH Periods
PUCCH receive channel adaptation	Beam resource utilization	This function adaptively adjusts the allocation of PUCCH receive channels based on the coverage and channel quality of UEs.	4.1.4 PUCCH Receive Channel Adaptation
Uplink IBLER adaptation optimization	System link performance	This function enables the gNodeB to determine the channel quality change for individual UEs based on the standard deviation of signal to interference plus noise ratio (SINR) fluctuations. This way, the gNodeB can obtain more accurate information about channel quality changes for UEs and select a more appropriate target initial block error rate (IBLER) accordingly.	4.1.5 Uplink IBLER Adaptation Optimization

4.1.1 Uplink Packet Bundled Scheduling

When a cell has medium or heavy load in the uplink (for example, when uplink PRB usage is higher than 60%), beam resources usually become insufficient earlier than PRB resources. Therefore, when UEs running small-packet services occupy beam resources exclusively, PUSCH PRBs may not be fully utilized due to beam resource insufficiency. To address this issue, uplink packet bundled scheduling is introduced. This function is controlled by the **UL_PACKET_AGGREGATED_SCH_SW** option of the **NRDUCellUISch.UISchAlgoSwitch** parameter. After this function takes effect, as shown in [Figure 4-1](#), **the scheduling occasion for the UE running large-packet services is adaptively adjusted, enabling beam resources to be fully utilized and thereby improving uplink PRB usage**. This function applies only to 4T4R or 8T8R cells.

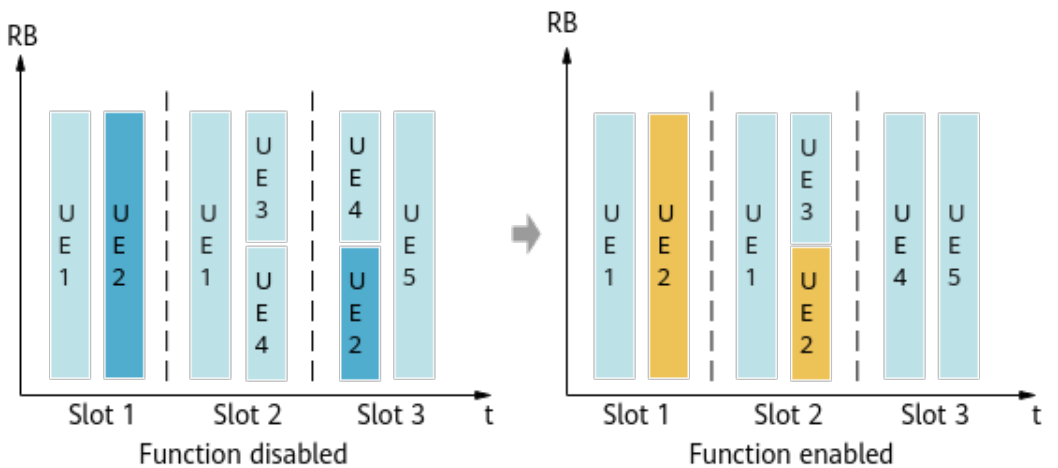
Figure 4-1 Schematic diagram of uplink packet bundled scheduling



4.1.2 Packet-Length-based Scheduling Optimization

In cells with medium or heavy load in the uplink (for example, with uplink PRB usage being higher than 60%), UEs' varied signal quality and service models make it possible to reduce the total transmission delay of UEs. To achieve this, packet-length-based scheduling optimization is introduced. This function is controlled by the **PKT_LEN_BASED_SCH_OPT_SW** option of the **NRDUCellPusch.PuschPerformanceSwitch** parameter. **Packet-length-based scheduling optimization adaptively selects UEs to schedule based on their service models and signal quality, with the aim to minimize total transmission delay of UEs, thereby increasing the average uplink UE throughput.** After this function is enabled, small-packet-service UEs with smaller volumes of buffered data and better signal quality are preferentially scheduled. As shown in [Figure 4-2](#), some data originally scheduled for UE 2 in slot 3 is now scheduled in slot 2, reducing the scheduling duration (that is, the transmission delay) of UE 2 by one slot.

Figure 4-2 Schematic diagram of packet-length-based scheduling optimization



4.1.3 Dual PRACH Periods

In high-frequency cells, PRACH resource utilization needs to be optimized to improve user performance and enhance the resource utilization policy. The frame

number and slot number of a preamble sent on the PRACH are determined by the PRACH configuration index. For details, see section 6.3.3 "Physical random-access channel" in 3GPP TS 38.211 V17.4.0. The "dual PRACH periods" function extends the PRACH period for CBRA and this allows the number of slots occupied by the PRACH within a given time segment to be reduced, thereby increasing the number of slots available for the PUSCH. In addition, this function enables the base station to receive preambles for CFRA in specific slots, further reducing the amount of time-frequency resources occupied by the PRACH.

The "dual PRACH periods" function allows the PRACH period for CBRA and that for CFRA to be separately configured using the **NRDUCellPrach.PrachConfigurationIndex** and **NRDUCellPrach.CfraPrachConfigIndex** parameters, respectively, only under identical settings of the preamble format, PUSCH subcarrier spacing (SCS), and slot configuration. Table 4-2 describes the detailed configurations for the function activation in the current version.

Table 4-2 Configurations for dual PRACH periods

Slot Configuration	PUSCH SCS	Short Preamble Format ^a	Configurations for Dual PRACH Periods
4_1_DDD SU	120 kHz	C2	- The NRDUCellPrach.PrachConfigurationIndex parameter is set to 192 to specify the PRACH period for CBRA. - The NRDUCellPrach.CfraPrachConfigIndex parameter is set to 188 to specify the PRACH period for CFRA.
4_1_DDD SU	120 kHz	A3	- The NRDUCellPrach.PrachConfigurationIndex parameter is set to 81 to specify the PRACH period for CBRA. - The NRDUCellPrach.CfraPrachConfigIndex parameter is set to 74 to specify the PRACH period for CFRA.
a: For details about short preamble formats, see <i>Random Access Control</i> .			

4.1.4 PUCCH Receive Channel Adaptation

The PUCCH receive channel adaptation function adaptively adjusts the allocation of PUCCH receive channels based on the coverage and channel quality of UEs. The **NRDUCellPucch.PucchRxAllocMode** parameter specifies whether this function is enabled.

- When this parameter is set to **ADAPTIVE**, this function is enabled, and the number of PUCCH receive channels is adaptively adjusted based on the coverage and channel quality of UEs. For example, fewer PUCCH receive

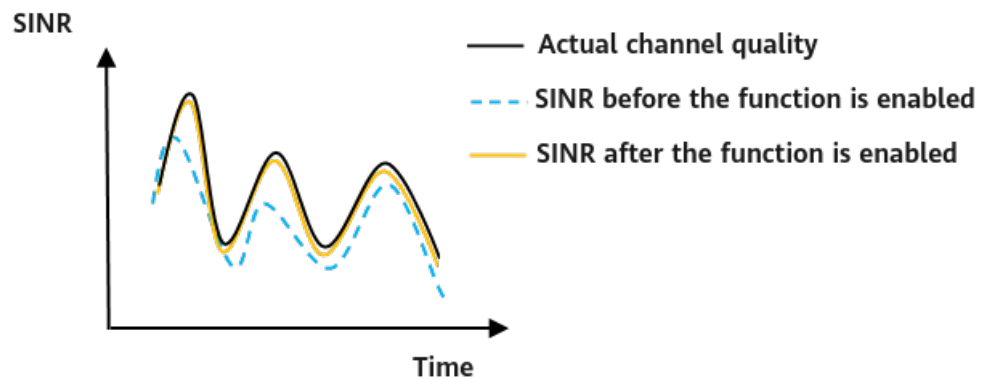
channels are used for cell-center UEs with good signal quality than for other UEs, reducing uplink resource consumption. Conversely, more PUCCH receive channels are used for cell-edge UEs with poor signal quality, improving user experience.

- When this parameter is set to **OFF**, this function is disabled and the receive channel allocation mode is the same for all UEs.

4.1.5 Uplink IBLER Adaptation Optimization

A fixed target IBLER cannot always match the need for signal to interference plus noise ratio (SINR) correction. As a result, the channel quality indicated by the SINR used for modulation and coding scheme (MCS) selection is significantly different from the actual channel quality and this restricts throughput. To select a more appropriate target IBLER and thus enable UEs to have higher uplink throughput, uplink IBLER adaptation optimization is introduced. This function is controlled by the **UL_IBLER_ADAPT_OPT_SW** option of the **NRDUCellUIMc.UlAmcAlgoSw** parameter. **After uplink IBLER adaptation optimization takes effect, the gNodeB determines the channel quality change for individual UEs based on the standard deviation of SINR fluctuations. This way, the gNodeB can obtain more accurate information about channel quality changes for UEs and select a more appropriate target IBLER accordingly, as illustrated in Figure 4-3.**

Figure 4-3 Schematic diagram of uplink IBLER adaptation optimization



After uplink IBLER adaptation optimization is enabled, the following parameters can be used to set the thresholds for the adaptation:

- **NRDUCellUIMc.UlMcsIdxForLowIblerTarget**. This parameter specifies the MCS index threshold for the low target IBLER to take effect in the uplink. When the MCS index is greater than or equal to the value of this parameter, the low target IBLER takes effect.
- **NRDUCellUIMc.UlSinrAvgLowThld**. This parameter specifies the low threshold expressed as the average uplink SRS SINR. If the average uplink SINR of a UE is greater than this threshold and less than or equal to the SINR value corresponding to **NRDUCellUIMc.UlMcsIdxForLowIblerTarget**, the UE is considered to be of slightly poor channel quality in terms of the average SINR. If the average uplink SINR of a UE is less than or equal to this threshold, the UE is considered to be of poor channel quality in terms of the average SINR.

- **NRDUCellUplinkAmdc.ULSnrAvgHighThld**. This parameter specifies the high threshold expressed as the average uplink SRS SINR. If the average uplink SINR of a UE is greater than this threshold, the UE is considered to be of good channel quality in terms of the average SINR. If the average uplink SINR of a UE is less than or equal to this threshold and greater than the SINR value corresponding to **NRDUCellUplinkAmdc.ULMcsIdxForLowblerTarget**, the UE is considered to be of slightly good channel quality in terms of the average SINR.
- **NRDUCellUplinkAmdc.ULSnrDiffLowThld**. This parameter specifies the lower threshold of the uplink SRS SINR standard deviation. If the uplink SINR standard deviation of a UE is less than or equal to this threshold, the UE is considered to be of high channel quality in terms of SINR standard deviation. If the uplink SINR standard deviation of a UE is greater than this threshold and less than or equal to **NRDUCellUplinkAmdc.ULSnrDiffHighThld**, the UE is considered to be of medium channel quality in terms of SINR standard deviation.
- **NRDUCellUplinkAmdc.ULSnrDiffHighThld**. This parameter specifies the upper threshold of the uplink SRS SINR standard deviation. If the uplink SINR standard deviation of a UE is greater than this threshold, the UE is considered to be of poor channel quality in terms of SINR standard deviation. If the uplink SINR standard deviation of a UE is less than or equal to this threshold and greater than **NRDUCellUplinkAmdc.ULSnrDiffLowThld**, the UE is considered to be of medium channel quality in terms of SINR standard deviation.

4.2 Network Analysis

4.2.1 Benefits

After Uplink IAS takes effect, **Cell Uplink Average Throughput (DU)** and **User Uplink Average Throughput (DU)** increase. The specific benefits depend on factors such as the traffic model and network load.

Table 4-3 describes the benefits of each sub-function.

Table 4-3 Benefits of each sub-function

Sub-function Name	Benefits
Packet-length-based scheduling optimization	In medium- or heavy-load scenarios (for example, uplink PRB usage is higher than 60%), packet-length-based scheduling optimization can shorten the service waiting time of UEs, and increase average uplink UE throughput while keeping Cell Uplink Average Throughput (DU) almost unchanged. When there are multiple high-frequency cells, it is recommended that the average uplink UE throughput of all cells in a sector be checked. For example, a sector has two cells: cell 1 and cell 2. Assume that the values of N.ThpVol.UL, N.ThpVol.UE.UL.SmallPkt, and N.ThpTime.UE.UL.RmvSmallPkt of cell 1 are a1, b1, and c1, respectively, and the values of N.ThpVol.UL, N.ThpVol.UE.UL.SmallPkt, and N.ThpTime.UE.UL.RmvSmallPkt of cell 2 are a2, b2, and c2, respectively. Check the average value of the two cells, that is, result of $[(a1 - b1) + (a2 - b2)]/(c1+c2)$.
Uplink packet bundled scheduling	In medium- or heavy-load scenarios (for example, uplink PRB usage is higher than 60%), uplink packet bundled scheduling increases uplink PRB usage and Cell Uplink Average Throughput (DU).
Dual PRACH periods	In heavy-load scenarios (for example, uplink PRB usage is higher than 75%), the "dual PRACH periods" function reduces PRACH resource consumption, increases the amount of available PUSCH resources, and therefore increases User Uplink Average Throughput (DU).
PUCCH receive channel adaptation	PUCCH receive channel adaptation increases Cell Uplink Average Throughput (DU) and User Uplink Average Throughput (DU). More cell-center UEs and higher uplink PRB usage can lead to larger increases.
Uplink IBLER adaptation optimization	Uplink IBLER adaptation optimization increases User Uplink Average Throughput (DU) when there are some cell-center UEs and cell-edge UEs (the average uplink MCS index for cell-center UEs ranges from 20 to 27 and that for cell-edge UEs ranges from 0 to 6).

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

After Uplink IAS takes effect, there are fluctuations in **Cell Downlink Average Throughput (DU)** and **User Downlink Average Throughput (DU)**.

[Table 4-4](#) describes the impacts of each sub-function.

Table 4-4 Network impacts of the sub-functions

Sub-function Name	Impact on the Network
Packet-length-based scheduling optimization	When there is a large number of UEs running small-packet services in a cell, packet-length-based scheduling optimization increases the probability that these UEs are scheduled in advance. As a result, uplink PRB usage and Cell Uplink Average Throughput (DU) decrease. In addition, after this function takes effect, if the number of uplink scheduling occasions increases, the number of downlink response packets increases. This may cause Cell Downlink Average Throughput (DU) and User Downlink Average Throughput (DU) to decrease.
Uplink packet bundled scheduling	Uplink packet bundled scheduling causes fluctuations in User Uplink Average Throughput (DU) . These fluctuations vary depending on the carrier configuration and inbound traffic model of UEs.

Sub-function Name	Impact on the Network
Dual PRACH periods	This function requires that the related parameters be set to identical values for NR cells that are served by the same RF module. Otherwise, UE access failures or service drops may occur. After this function takes effect in heavy-load scenarios (for example, uplink PRB usage is higher than 75%), there will be fluctuations in the values of the following indicators: • Cell Uplink Average Throughput (DU) and Cell Downlink Average Throughput (DU) • Uplink PRB usage (N.PRB.Ul.Used.Avg divided by $\text{N.PRB.Ul.Avail.Avg}$) and downlink PRB usage (Downlink Resource Block Utilizing Rate (DU)) • Number of online UEs in a cell • Average downlink MCS index and average uplink MCS index - Average DownLink MCS=$\frac{(0 \times \text{N.ChMeas.PDSCH.MCS.0} + 1 \times \text{N.ChMeas.PDSCH.MCS.1} + 2 \times \text{N.ChMeas.PDSCH.MCS.2} + \dots + 28 \times \text{N.ChMeas.PDSCH.MCS.28})}{(\text{N.ChMeas.PDSCH.MCS.0} + \text{N.ChMeas.PDSCH.MCS.1} + \text{N.ChMeas.PDSCH.MCS.2} + \dots + \text{N.ChMeas.PDSCH.MCS.28})}$ - Average UpLink MCS=$\frac{(0 \times \text{N.ChMeas.PUSCH.MCS.0} + 1 \times \text{N.ChMeas.PUSCH.MCS.1} + 2 \times \text{N.ChMeas.PUSCH.MCS.2} + \dots + 28 \times \text{N.ChMeas.PUSCH.MCS.28})}{(\text{N.ChMeas.PUSCH.MCS.0} + \text{N.ChMeas.PUSCH.MCS.1} + \text{N.ChMeas.PUSCH.MCS.2} + \dots + \text{N.ChMeas.PUSCH.MCS.28})}$ • Average downlink rank and average uplink rank - Average DownLink RANK=$\frac{(1 \times \text{N.PDSCH.InitTbDL.Rank1} + 2 \times \text{N.PDSCH.InitTbDL.Rank2} + 3 \times \text{N.PDSCH.InitTbDL.Rank3} + 4 \times \text{N.PDSCH.InitTbDL.Rank4})}{(\text{N.PDSCH.InitTbDL.Rank1} + \text{N.PDSCH.InitTbDL.Rank2} + \text{N.PDSCH.InitTbDL.Rank3} + \text{N.PDSCH.InitTbDL.Rank4})}$ - Average UpLink RANK=$\frac{(1 \times \text{N.PUSCH.InitTbUL.Rank1} + 2 \times \text{N.PUSCH.InitTbUL.Rank2})}{(\text{N.PUSCH.InitTbUL.Rank1} + \text{N.PUSCH.InitTbUL.Rank2})}$ • Proportions of uplink and downlink scheduling occasions using different modulation schemes There will be the following changes in the values of access delay indicators: • The delay for the IDLE2ACTIVE state transition increases. • The attach delay increases.

Sub-function Name	Impact on the Network
PUCCH receive channel adaptation	After PUCCH receive channel adaptation takes effect, Cell Uplink Average Throughput (DU) and User Uplink Average Throughput (DU) increase; however, the number of downlink radio link control (RLC) status reports or Transmission Control Protocol (TCP) response packets increases, which results in fluctuations in Cell Downlink Average Throughput (DU) and User Downlink Average Throughput (DU) .
Uplink IBLER adaptation optimization	In cells with heavy downlink load (for example, downlink PRB usage is higher than 70%) and a high proportion of cell-edge UEs, uplink IBLER adaptation optimization enables UEs to provide more timely feedback on downlink data transmission. This may increase downlink traffic at the Packet Data Convergence Protocol (PDCP) layer. However, User Downlink Average Throughput (DU) may decrease due to limited downlink air interface capabilities.

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-101241	Uplink IAS	NR0S00PCDW00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	FR2 uplink experience improvement	FR2_EXP_IMP_SW option of the NRDUCellPuschPerformanceSwitch parameter	<i>Intelligent Aggregate Scheduling (High-Frequency TDD)</i>	- FR2 uplink experience improvement must be enabled before the "dual PRACH periods" and PUCCH receive channel adaptation functions can be enabled. - Uplink packet bundled scheduling, packet-length-based scheduling optimization, and uplink IBLER adaptation optimization can take effect only after FR2 uplink experience improvement is enabled.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	PRACH Configuration Index	NRDUCellPrach.PrachConfigurationIndex	<i>Random Access Control</i>	• The NRDUCellPrach.CfraPrachConfigurationIndex parameter can be set to 188 only when the NRDUCellPrach.PrachConfigurationIndex parameter is set to 192. • The NRDUCellPrach.CfraPrachConfigurationIndex parameter can be set to 74 only when the NRDUCellPrach.PrachConfigurationIndex parameter is set to 81. • In other cases, the NRDUCellPrach.CfraPrachConfigurationIndex parameter must be set to 65535.
High-frequency TDD	PUCCH symbol number adaptation	PUCCH_RES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter	<i>Channel Management</i>	PUCCH receive channel adaptation can take effect only after PUCCH symbol number adaptation is enabled.

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave 3D coverage pattern	NRDUCellTrpMmwavBeam.CoverageScenario	<i>mmWave Beam Management (High-Frequency TDD)</i>	PUCCH receive channel adaptation cannot be enabled when the coverage scenario is single-beam coverage (that is, when the NRDUCellTrpMmwavBeam.CoverageScenario parameter is set to SCENARIO_103).
High-frequency TDD	Increased downlink symbol proportion in self-contained slots	S_SLOT_DL_SYM_RATIO_INC_SW option of the NRDUCellDL-SchRes.DLSchResAlgoSw parameter	None	PUCCH receive channel adaptation cannot be enabled together with the "increased downlink symbol proportion in self-contained slots" function.

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Maximum Preamble Transmission Count	NRDUCellPrach.MaxPreambleTransC nt	<i>Random Access Control</i>	If the "dual PRACH periods" function is enabled and the Maximum Preamble Transmission Count parameter is set to a value less than the scenario-specific recommended values, there is a low probability that UEs experience service drops. The recommended values are as follows: • N50 if the FLEXIBLE_DENSE_BEAM_SW option of the NRDUCellTrpMmwavBeam.BeamPerformanceSw parameter is selected • N20 if the FLEXIBLE_DENSE_BEAM_SW option of the NRDUCellTrpMmwavBeam.BeamPerformanceSw parameter is deselected

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	PUCCH grouping adaptation enhancement for inter-FR CA	PUCCH_GROUP_ADAPT_ENH_SW option of the NRDUCellCarrierMgmt.InterFrCaEnhSw parameter	<i>Carrier Aggregation</i>	PUCCH receive channel adaptation does not take effect for UEs with PUCCH grouping adaptation enhancement for inter-FR CA in effect.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR TDD-capable main control boards and baseband processing units that work in mmWave frequency bands support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF modules

- Uplink packet bundled scheduling
The RF module must be a 4T4R or 8T8R AAU that supports NR TDD high frequency bands.
- PUCCH receive channel adaptation
The RF module must be the HAAU5233w.
- Other sub-functions
All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Cells

Uplink packet bundled scheduling requires that the **NRDUCellTrp.TxRxMode** parameter be set to **4T4R** or **8T8R**.

The "dual PRACH periods" and PUCCH receive channel adaptation functions require that the **NRDUCell.SlotAssignment** parameter be set to **4_1_DDDSU** for NR cells.

The PUCCH receive channel adaptation function requires that the **NRDUCell.NrDuCellNetworkingMode** parameter be set to a value other than **HYPER_CELL**.

If the parameter for the PUCCH receive channel adaptation function is set to different values for cells in a sector, this function may not take effect in some or all of these cells.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-5 describes the required parameters.

Table 4-5 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
PUSCH Performance Switch	NRDUCellPusch.PuschPerformanceSwitch	FR2_EXP_IMP_SW	Select this option if functions for FR2 uplink experience improvement are required.
UL Scheduling Algorithm Switch	NRDUCellULSch.UlSchAlgorithmSwitch	UL_PACKET_AGGREGATED_SCH_SW	In FWA scenarios with medium or heavy load (for example, with the uplink PRB usage being higher than 60%), it is recommended that this option be selected. In other scenarios, it is recommended that this option be deselected.
PUSCH Performance Switch	NRDUCellPusch.PuschPerformanceSwitch	PKT_LEN_BASED_SCH_OPT_SW	In FWA scenarios with medium or heavy load (for example, with the uplink PRB usage being higher than 60%), it is recommended that this option be selected. In other scenarios, it is recommended that this option be deselected.

Parameter Name	Parameter ID	Option	Setting Notes
CFRA PRACH Configuration Index	NRDUCellPrach.CfraPrachConfigIndex	None	The default value of this parameter is 65535 , which indicates that the "dual PRACH periods" function is disabled. If this parameter is set to 74 or 188 , the "dual PRACH periods" function is enabled.
PRACH Configuration Index	NRDUCellPrach.PrachConfigurationIndex	None	Recommended settings for dual PRACH periods: • 81 when the NRDUCellPrach.CfraPrachConfigIndex parameter is set to 74 • 192 when the NRDUCellPrach.CfraPrachConfigIndex parameter is set to 188 For the settings in other scenarios, see <i>Random Access Control</i> .
Maximum Preamble Transmission Count	NRDUCellPrach.MaxPreambleTransCnt	None	Recommended settings for dual PRACH periods: • N50 if the FLEXIBLE_DENSE_BEAM_SW option of the NRDUCellTrpMmwav-Beam.BeamPerformanceSw parameter is selected • N20 if the FLEXIBLE_DENSE_BEAM_SW option of the NRDUCellTrpMmwav-Beam.BeamPerformanceSw parameter is deselected
PUCCH RX Allocation Mode	NRDUCellPucch.PucchRxAllocMode	None	Use the recommended value.
Uplink AMC Algorithm Switch	NRDUCellUlAmc.UlAmcAlgoSw	UL_IBLER_ADA PT_OPT_SW	It is recommended that this option be selected when the uplink PRB usage is higher than 30%.

Parameter Name	Parameter ID	Option	Setting Notes
UL SINR Average Low Threshold	NRDUCellULAmc.UlSinrAvgLowThld	None	Use the recommended value.
UL SINR Average High Threshold	NRDUCellULAmc.UlSinrAvgHighThld	None	Use the recommended value.
UL SINR Difference Low Threshold	NRDUCellULAmc.UlSinrDiffLowThld	None	Use the recommended value.
UL SINR Difference High Threshold	NRDUCellULAmc.UlSinrDiffHighThld	None	Use the recommended value.
UL MCS Index Thld for Low IBLER Target	NRDUCellULAmc.UlMcsIdxForLowBlrTarget	None	Use the recommended value.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling FR2 uplink experience improvement
MOD NRDUCCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch = FR2_EXP_IMP_SW-1;
//Enabling uplink packet bundled scheduling
MOD NRDUCCELLULSCH: NrDuCellId=0, UlschAlgoSwitch=UL_PACKET_AGGREGATED_SCH_SW-1;
//Enabling packet-length-based scheduling optimization
MOD NRDUCCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=PKT_LEN_BASED_SCH_OPT_SW-1;
//Enabling dual PRACH periods (using 50 as the maximum number of preamble transmissions for example)
MOD NRDUCCELLPRACH: NrDuCellId=0, PrachConfigurationIndex=192, CfraPrachConfigIndex=188,
MaxPreambleTransCnt=N50;
MOD NRDUCCELLPRACH: NrDuCellId=0, PrachConfigurationIndex=81, CfraPrachConfigIndex=74,
MaxPreambleTransCnt=N50;
//Enabling PUCCH receive channel adaptation
MOD NRDUCCELLPUCCH: NrDuCellId=0, PucchRxAllocMode=ADAPTIVE;
//Enabling uplink IBLER adaptation optimization and setting the corresponding thresholds
MOD NRDUCCELLULAMC: NrDuCellId=0, UIAmcAlgoSw=UL_IBLER_ADAPT_OPT_SW-1,
```

```
UISnrDiffLowThld=121, UISnrDiffHighThld=189, UIMcsIdxForLowIblerTarget=13, UISnrAvgLowThld=100,  
UISnrAvgHighThld=2240;
```

Deactivation Command Examples

```
//Disabling uplink packet bundled scheduling  
MOD NRDUCELLULSCH: NrDuCellId=0, ULSchAlgoSwitch=UL_PACKET_AGGREGATED_SCH_SW-0;  
//Disabling packet-length-based scheduling optimization  
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=PKT_LEN_BASED_SCH_OPT_SW-0;  
//Disabling dual PRACH periods  
MOD NRDUCELLPRACH: NrDuCellId=0, CfraPrachConfigIndex=65535;  
//Disabling PUCCH receive channel adaptation  
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchRxAllocMode=OFF;  
  
//Disabling uplink IBLER adaptation optimization  
MOD NRDUCELLULAMC: NrDuCellId=0, ULAmcAlgoSw=UL_IBLER_ADAPT_OPT_SW-0;  
//Disabling FR2 uplink experience improvement  
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch = FR2_EXP_IMP_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Check whether the functions have taken effect based on uplink average cell throughput and uplink average UE throughput, which are indicated by the following indicators, respectively:

- **Cell Uplink Average Throughput (DU)** = $N.ThpVol.UL.Cell / N.ThpTime.UL.Cell$
- **User Uplink Average Throughput (DU)** = $(N.ThpVol.UL - N.ThpVol.UE.UL.SmallPkt) / N.ThpTime.UE.UL.RmvSmallPkt$

4.4.3 Network Monitoring

Monitor the performance of the feature by monitoring the performance counters listed in [Table 4-6](#).

Table 4-6 Basic performance counter monitoring

Counter Description	Counter Name
Volume of uplink data sent at the MAC layer in a cell	N.ThpVol.UL.Cell and N.ThpTime.UL.Cell
Volume of uplink data sent at the RLC layer in a cell	N.ThpVol.UL , N.ThpTime.UL , and N.ThpVol.UE.UL.SmallPkt
Number of times a specific MCS index is selected for downlink scheduling	N.ChMeas.PDSCH.MCS.0 to N.ChMeas.PDSCH.MCS.31

Counter Description	Counter Name
Number of times a specific MCS index is selected for uplink scheduling	N.ChMeas.PUSCH.MCS.0 to N.ChMeas.PUSCH.MCS.31
Number of RBs scheduled in the downlink	N.PRB.DL.Used.Avg and N.PRB.DL.Used.Avg
Number of RBs scheduled in the uplink	N.PRB.UL.Used.Avg and N.PRB.UL.Avail.Avg
Number of times a specific rank is used in downlink scheduling	N.PDSCH.InitTbDL.Rank1 to N.PDSCH.InitTbDL.Rank4

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 38.211: "NR; Physical channels and modulation"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *Massive MIMO AHR (Low-Frequency TDD)*
- *License Control Item Lists*
- *Massive MIMO Uplink Boosting (Low-Frequency TDD)*
- *Random Access Control*
- *mmWave Beam Management (High-Frequency TDD)*
- *Hyper Cell*
- *Channel Management*
- *Carrier Aggregation*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*